Composition of fuzzy set

$$A = [a_{ij}]$$

$$B = [b_{jk}]$$

$$C = A \circ B$$

$$[c_{ik}] = [a_{ij}] \circ [b_{jk}]$$

$$c_{ik} = \max\left[\min(a_{ij}, b_{jk})\right]$$

$$A = [a_{ij}] = \begin{bmatrix} 0.2 & 0.3 \\ 0.5 & 0.7 \end{bmatrix} \quad B = [b_{jk}] = \begin{bmatrix} 0.3 & 0.6 & 0.7 \\ 0.1 & 0.8 & 0.6 \end{bmatrix}$$

$$c_{11} = \max\left[\min(a_{11}, b_{11}), \min(a_{12}, b_{21})\right]$$

$$= \max\left[\min(0.2, 0.3), \min(0.3, 0.1)\right]$$

$$= 0.2$$

$$A = [a_{ij}] = \begin{bmatrix} 0.2 & 0.3 \\ 0.5 & 0.7 \end{bmatrix} \quad B = [b_{jk}] = \begin{bmatrix} 0.3 & 0.6 & 0.7 \\ 0.1 & 0.8 & 0.6 \end{bmatrix}$$

$$c_{12} = \max\left[\min(a_{11}, b_{12}), \min(a_{12}, b_{22})\right]$$
$$= \max\left[\min(0.2, 0.6), \min(0.3, 0.8)\right]$$
$$= 0.3$$

$$c_{13} = \max\left[\min(0.2, 0.7), \min(0.3, 0.6)\right]$$
$$= 0.3$$

$$c_{21} = \max\left[\min(0.5, 0.3), \min(0.7, 0.1)\right]$$
$$= 0.3$$

$$c_{22} = \max\left[\min(a_{21}, b_{12}), \min(a_{22}, b_{22})\right]$$
$$= \max\left[\min(0.5, 0.6), \min(0.7, 0.8)\right]$$
$$= 0.7$$

$$c_{23} = \max\left[\min(0.5, 0.7), \min(0.7, 0.6)\right]$$
$$= 0.6$$

$$C = \begin{bmatrix} 0.2 & 0.3 & 0.3 \\ 0.3 & 0.7 & 0.6 \end{bmatrix}$$

Properties of fuzzy sets.

1. Law of excluded middle:

In crisp set, $A \cup \bar{A} = X$

However, according to the fuzzy set

$$A \cup \bar{A} \neq X$$

2. Law of contridiction.

As per crisp set, $A \cap \bar{A} = \phi$

However in fuzzy set $A \cap \bar{A} \neq \phi$

Measurement of fuzziness and inaccuracy of fuzzy sets :

$$X = \{x_1, x_2, x_3 \ldots x_n\}$$

$A(x)$

H(A) is the entrophy of the fuzzy set A(x)

$$H(A) = \frac{-1}{n} \sum_{i=1}^{n} \left[\mu_A(x_i) \log\{\mu_A(x_i)\} + \{1-\mu_A(x_i)\} \log\{1-\mu_A(x_i)\} \right]$$

Ex:

$$A(x) = \{(x_1, 0.1), (x_2, 0.3), (x_3, 0.4), (x_4, 0.5)\}$$

$$H(A) = -\frac{1}{4} \Big[\{0.1 \times \log(0.1) + 0.9 \times \log(0.9)\} + \{0.3 \log(0.3) + 0.7 \times \log(0.7)\}$$
$$\{0.4 \times \log(0.4) + 0.6 \times \log(0.6)\} + \{0.5 \times \log(0.5) + 0.5 \times \log(0.5)\} \Big]$$

$$= 0.2499$$

Inaccuracy:

A(x) and B(x) are fuzzy sets.

$$I(A;B) = \frac{-1}{n}\sum_{i=1}^{n}\left[\mu_A(x_i)\log(\mu_B(x_i)) + (1-\mu_A(x_i)\log(1-\mu_B(x_i))\right]$$

Ex:

$A(x) = \{(x_1, 0.1), (x_2, 0.2), (x_3, 0.3), (x_4, 0.4)\}$

$B(x) = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0.8), (x_4, 0.9)\}$

Determine inaccuracy of the fuzzy set B(x) with respect to fuzzy set A(x).

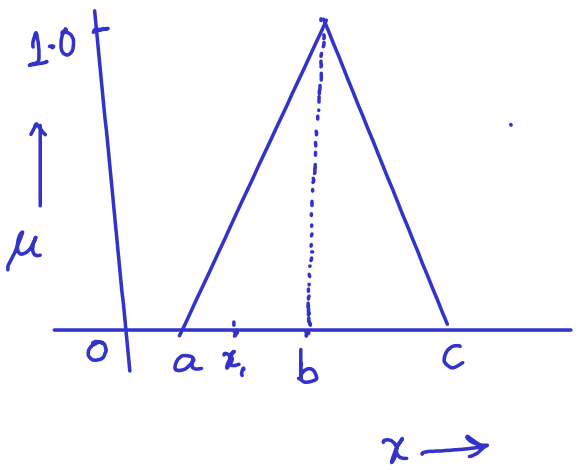
$$I(A;B) = \frac{-1}{4}\Big[\{0.1\times\log(0.5) + 0.9\times\log(0.5)\} + \{0.2\times\log(0.7) + (0.8)\times\log(0.3)\} + \{0.3\times\log(0.8) + 0.7\times\log(0.2)\} + \{0.4\times\log(0.9) + 0.6\times\log(0.1)\}\Big]$$

$$= \underline{0.4717}$$

Triangular membership:

It is denoted Triangle $(x; a, b, c)$

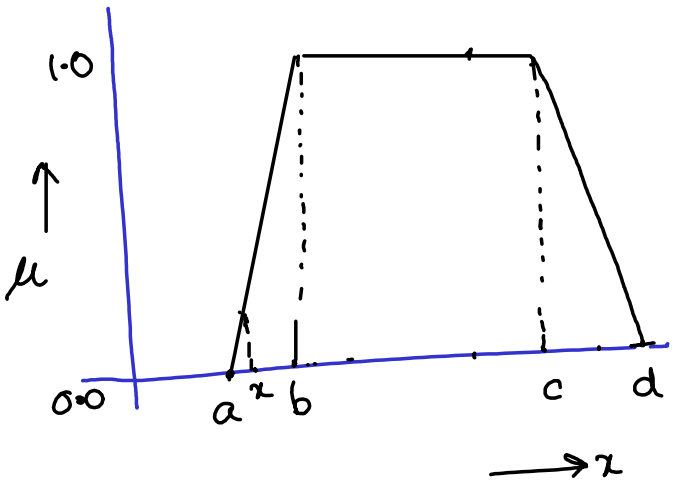
$\mu_{triangle}$



$$\mu_{triangle} = \max\left(\min\left(\frac{x-a}{b-a}, \frac{c-x}{c-b}\right), 0\right)$$

Trapezoidal Membership

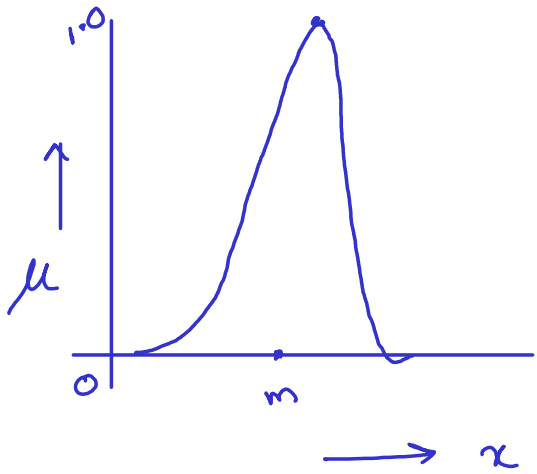
Trapezoidal$(x; a, b, c, d)$



$$\mu_{trapezoidal} = \max\left(\min\left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right), 0\right)$$

Gaussian Membership

Gaussian $(x; m, \sigma)$



$$\mu_{Gaussian} = \frac{1}{e^{\frac{1}{2}\left(\frac{x-m}{\sigma}\right)^2}}$$